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Figure 9. Snapshot of the visual diversity in the Matterport3D dataset, illustrating one randomly selected panoramic viewpoint per scene.

Instructions: Give A Smart Robot Directions (Click to collapse)

You will see a series of panoramic photos taken while moving from a **start location** to a **goal location** in a building. **Your task is to write directions so that a smart robot can find the goal location after starting from the same start location.** The robot understands language and recognizes objects about as well as a typical person. However, you should assume that the robot is visiting this building for the first time.

For your reference, the path to the goal is indicated by color-coded markers (**green** for start, **red** for goal, and **blue** for intermediate markers).

- You won't see the **green** start marker at the beginning - because it's under your feet.
- You may not see the **red** goal marker until you move (often the goal is in the next room).
- **These markers are not visible to the robot**, and should not be mentioned in your directions.

Good directions will ensure that the robot arrives **within a few metres** of the red goal marker. Therefore, we suggest:

- **NEW! Spelling and punctuation is important.** Please use full sentences with punctuation (,) and correct spelling.
- **Focus on the goal, not the path.** It's not necessary for the robot to follow the exact path indicated by the markers.
- **Try to mention objects or landmarks.** This is clearer than saying 'turn slight left' or 'go forward'.

Mouse Controls:

1. **Left-click and drag the panoramic image** to look around.
2. **Right-click on a color-coded marker** to move to that position.
3. **Press the 'Play / Replay' button** at any time to watch a 15-20 second animated fly-through from the start to the goal.

Before you start, [please watch this short training video](#). It contains examples that will help you complete these tasks efficiently.

Note: This task is not suitable for devices with small screens or touch screen devices. Recommended browsers are Chrome, Firefox and Safari (not Internet Explorer).

These tasks relate to academic research conducted by Peter Anderson through the [Australian Centre for Robotic Vision](#), Brisbane, Australia. We estimate that on average each HIT to take around 1-1.5 minutes to complete. Please send your queries and feedback to bringmeaspoon@gmail.com. We will be continually releasing more HITs for this task.



Left-click and drag the panoramic image to start.
Instructions have been updated from the first batch
(please re-read).

Play / Replay

Write your Directions here (with correct spelling and punctuation):

Submit

Figure 10. AMT data collection interface for the R2R navigation dataset. Here, blue markers can be seen indicating the trajectory to the goal location. However, in many cases the worker must first look around (pan and tilt) to find the markers. Clicking on a marker moves the camera to that location. Workers can also watch a 'fly-through' of the complete trajectory by clicking the Play / Replay button.